Chhaya — Open-Source License & Dataset Details

Complete Software IP, Clinical Dataset Governance, Font Software & Stack Manifest

Team SynapseX · D.B.S Kanpur

Happiest Health Digital Twin Challenge 2026

Submission Folder: SynapseX_D.B.S Kanpur

1. Core Repository Code

Chhaya Software System

MIT LICENSE

License: MIT License (Copyright © 2026 The Chhaya authors · Team SynapseX, D.B.S Kanpur)

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Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the condition that the above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.
```

Covers: ODE physiological twin solver (`src/chhaya/twin/`), Bayesian Laplace calibration, baseline and ML models (`src/chhaya/models/`), Gates 1, 2, and 3 evaluation suites (`src/chhaya/eval/`), local clinician dashboard server (`src/chhaya/dashboard/`), and test suites (`tests/`).

2. Clinical Dataset Provenance & Zero-Redistribution Governance

DATASET	COHORT PROFILE	LICENSE	ORIGINAL AUTHORITY & DOI CITATION	CHECKSUM & STORAGE POLICY
ShanghaiT2DM	100 inpatient Type 2 diabetes patients (109 wears), 15-min CGM readings, daily reference fingersticks, lab panels, diet logs.	CC BY 4.0 Attribution Required	Zhao et al., *Scientific Data* 2023. `doi:10.6084/m9.figshare.c.6310860`	MD5: `4bfb61cfa506b481...` Downloaded dynamically; not redistributed.
CGMacros	45 free-living individuals, dual continuous Dexcom & Abbott Libre sensors, Fitbit Charge 4 wristband, meal macronutrients.	CC BY-NC-SA 4.0 Non-Commercial Only	Das et al., PhysioNet 2024. `doi:10.13026/3z8q-x658`	SHA256: `05c8b0e6f1a27570...` Downloaded dynamically; not redistributed.

Governance Rule (Zero Redistribution)

Raw patient traces are never redistributed in the Git repository. They are fetched dynamically via automated checksum-verified download scripts (`python -m chhaya.data.download`) directly from authoritative archives to respect patient privacy and original license conditions. Per-reading time series remain local; only aggregate evaluation benchmarks and the synthetic patient are committed.

3. Typography & Offline Clinic Font Software

Adobe Source Font Family (Source Sans 3, Source Serif 4, Source Code Pro)

SIL OFL 1.1

Bundled locally in `src/chhaya/dashboard/static/fonts/` for fully air-gapped hospital operation without external CDN dependencies or tracking scripts.

License: SIL Open Font License 1.1 (OFL-1.1) (Copyright © 2010–2024 Adobe Systems Incorporated). Full legal notice preserved in `src/chhaya/dashboard/static/fonts/OFL.txt`.

4. Third-Party Computational Stack

All runtime packages declared in `pyproject.toml` are governed by OSI-approved permissive licenses:

COMPONENT	LICENSE	ROLE IN CHHAYA ARCHITECTURE
JAX / jaxlib	APACHE 2.0	Vectorized numerical ODE integration and compiled autodiff Jacobian evaluations.
NumPy / SciPy	BSD 3-CLAUSE	Array algebra, L-BFGS-B MAP optimization, and Wilcoxon signed-rank hypothesis tests.
Pandas / Scikit-Learn	BSD 3-CLAUSE	Tabular clinical time series alignment, cross-validation, and AUPRC/AUROC scoring.
PyArrow / Requests	APACHE 2.0	Feather/Parquet serialization and checksummed HTTPS dataset ingestion.
LightGBM	MIT	Gradient boosted tree excursion classification baseline.
Hatchling / Tabulate / OpenPyXL	MIT	Build packaging system, CLI scorecard generation, and clinical Excel sheet parsing.

5. Front-End & Media Assets

Front-End, Audio & Synthetic Avatar Disclosure

NATIVE & MIT

- **Front-End Architecture:** 100% Native Vanilla HTML5, Vanilla CSS3, and dynamic SVG rendering (`src/chhaya/dashboard/static/app.js`). Zero external JavaScript frameworks (no React, Vue, Angular, D3, or Chart.js) and zero telemetry or analytics scripts.
- **Ambient Audio (`video/out/music.mp3`):** Royalty-free acoustic soundtrack utilized strictly for video demonstration and walkthrough mixing.
- **Demo Patient (`Mrs. R.`):** Synthetic clinical avatar (58 yr female, HbA1c 7.8 %, Fasting glucose 142 mg/dL, Metformin 1000 mg bid, BMI 28.4 kg/m²) with zero real-world patient identity, licensed under MIT terms.

6. Scientific & Physiological Literature Lineage

Open Peer-Reviewed Literature Foundations

OPEN SCIENCE

- **E-DES Glucose-Insulin Model:** Maas AH et al. *A physiology-based model describing heterogeneity in glucose metabolism: the core of the Eindhoven Diabetes Education Simulator (E-DES).* J Diabetes Sci Technol 2015;9:282-292. (PMC4604593).
- **Personalized CGM Parameter Identification:** Erdős B et al. *Leveraging continuous glucose monitoring for personalized modeling of insulin-regulated glucose metabolism.* Scientific Reports 2021.